

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD):

FR-DLT-001: Load Customer and Order Data to Delta Tables

Title: Load Customer and Order Data to Delta Tables

Description: The system must read source CSV data from volume and load it into delta tables for customer and order data.

Preconditions:

CSV files are available at the specified locations:

`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`

- Delta tables for customer and order data do not exist or are empty

Main Flow / Functional Steps:

1. Read customer CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and load it into a delta table named "customer".
2. Read order CSV data from `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` and load it into a delta table named "order".
3. Add a new column "TotalAmount" to the "order" delta table, calculated as the product of "PricePerUnit" and "Qty" columns.

FR-DLT-002: Clean and Process Customer and Order Data

Title: Clean and Process Customer and Order Data

Description: The system must remove null and duplicate records from customer and order delta tables.

Preconditions:

- Customer and order delta tables are loaded with data

Main Flow / Functional Steps:

1. Remove null records from both "customer" and "order" delta tables.
2. Remove duplicate records from both "customer" and "order" delta tables.

FR-DLT-003: Create and Load ordersummary Table

Title: Create and Load ordersummary Table

Description: The system must create an "ordersummary" table if it does not exist and load it with joined customer and order data using SCD type 2.

Preconditions:

- Customer and order delta tables are cleaned and processed
- "ordersummary" table does not exist or is empty

Main Flow / Functional Steps:

1. Create "ordersummary" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Join "customer" and "order" delta tables using "CustId" field.
3. Load the joined data into "ordersummary" table using SCD type 2.

FR-DLT-004: Update ordersummary Table on Customer Changes

Title: Update ordersummary Table on Customer Changes

Description: The system must update the "ordersummary" table whenever there is a change in the "customer" table.

Preconditions:

- "ordersummary" table is loaded with data
- "customer" table is updated

Main Flow / Functional Steps:

1. Detect changes in the "customer" table.
2. Update the "ordersummary" table by making old records inactive and new records active.
3. Update StartDate and EndDate accordingly to maintain history.

FR-DLT-005: Create and Load customeraggregatespend Table

Title: Create and Load customeraggregatespend Table

Description: The system must create a "customeraggregatespend" table if it does not exist and load it with aggregated data from "ordersummary" table.

Preconditions:

- "ordersummary" table is loaded with data

- "customeragggregatespend" table does not exist or is empty

Main Flow / Functional Steps:

1. Create "customeragggregatespend" table if it does not exist in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard".
2. Aggregate "TotalAmount" column from "ordersummary" table and group by "Name" and "Date" columns.
3. Load the aggregated data into "customeragggregatespend" table.

FR-DLT-006: Implement Delta Live Tables

Title: Implement Delta Live Tables

Description: The system must implement delta live tables for the entire workflow.

Preconditions:

- All previous steps are completed successfully

Main Flow / Functional Steps:

1. Implement delta live tables for loading customer and order data.
2. Implement delta live tables for cleaning and processing customer and order data.
3. Implement delta live tables for creating and loading "ordersummary" table.
4. Implement delta live tables for updating "ordersummary" table on customer changes.
5. Implement delta live tables for creating and loading "customeragggregatespend" table.